| NOT MEASUREMENT SENSITIVE |

MIL-T-81098F
20 February 1991

SUPERSEDING
MIL-T-81098E
31 October 1984

MILITARY SPECIFICATION

TRUCK, AMMUNITION HANDLING: STAKE AND PLATFORM BODY, 24,000 POUNDS GVW, 4 BY 2 OR 4 BY 4

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a commercial, gasoline or diesel engine-driven, four-wheel, two-rear-wheel-drive, or four-wheel-drive, stake and platform truck having a minimum gross vehicle weight (gvw) rating of 24,000 pounds (lbs), equipped with a rear-of-cab mounted revolving articulating boom type crane. The trucks procured under this specification are commercial items which shall be warranted as specified in the procurement documents.

1.2 Classification. Truck shall be one of the following types, as specified (see 6.2):

Type I - 4 by 2 truck chassis.
Type II - 4 by 4 truck chassis.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2320

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-393 - Extinguisher, Fire, Dry-Chemical (Hand-Portable).
W-B-131 - Battery, Storage: Vehicular, Ignition, lighting and Starting.
ZZ-T-381 - Tire, Pneumatic, Vehicular (Highway).

MILITARY

MIL-P-514 - Plate, Identification, Instruction And Marking, Blank.
MIL-E-6051 - Electromagnetic Compatibility Requirements, Systems.
MIL-V-62038- Vehicle, Wheeled, Preparation For Shipment And Storage Of.

STANDARDS

FEDERAL

FED-STD-297 - Rustproofing of Commercial (Nontactical Vehicles).

MILITARY

MIL-STD-1223 - Non-Tactical Wheeled Vehicles Treatment, Painting, Identification Marking, Data Plates Standards.
MIL-STD-1474 - Noise Limits for Army Materiel.
MS51317 - Light, Warning, Vehicular Rotating, DC.
MS51318 - Headlight, Blackout, 24 Volt, Waterproof.
MS51335 - Pintle Assembly, Towing, 18,000 lbs, Capacity, Manual Release.
MS52125 - Composite Light Tail, Stop, Turn and Marker.
MS52126 - Composite Light Front, Turn, Park and Marker.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines: Certification and Test Procedures.
Interstate Motor Carrier Noise Emission Standards.
Motor Vehicle Air Pollution Standards.
Title 40, CFR, Part 204 Subpart B - Noise Emission Standards for Construction Equipment.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852; or CFR, Title 40 should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Non-Governmemt publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which is current on the date of the solicitation (see 6.2)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI B30.5 - Safety Standards for Cableways Cranes, Derricks, Hoists, Hooks, Jacks and Slings Mobile and Locomotive Cranes;

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Standards and Recommended Practices

SAE J350 - Spark Arrestor Test Procedure for Medium Size Engines.
SAE J534 - Lubrication Fittings.
SAE J551 - Performance Levels and Methods of Measurements of Electromagnetic Radiation..
SAE J560 Seven-Conductor Electrical Connector for Truck-Trailer.

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SAE J682 - Rear Wheel Splash and Stone Throw Protection (DoD adopted).
SAE J683 - Tire Chains Clearance-Trucks, Buses, and Combinations of Vehicles.
SAE J688 - Truck Ability Prediction Procedure (DoD adopted).
SAE J704 - Openings for Six- and Eight-Bolt Truck Transmission Mounted Power Take-Offs.
SAE J706 - Ratings of Winches.
SAE J1063 - Cantilevered Boom Crane Structures-Method of Test.
SAE J1349 - Engine Power Test Code- Spark Ignition and Diesel.

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Suite 304, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The truck shall be a four-wheel, two-rear-wheel-drive, or four-wheel-drive gasoline or diesel engine-driven-truck cab-and-chassis having a minimum gvw rating of 24,000 lbs. A fully hydraulic, articulated boom type crane shall be frame mounted at rear of cab. A platform body shall be located at the rear of the crane. The platform body may be equipped, on an optional basis, with a cargo conversion unit consisting of removable type body front and side panels, and tailgate.

3.2 First production vehicle. The contractor shall furnish a truck for first production vehicle inspection.

* 3.3 Standard truck components and accessories. Except as specified in 3.3.1 through 3.3.1.15, the truck, components, and accessories shall be standard or optional items which meet or exceed the requirements of this specification. All items shall be represented and rated in the manufacturer's sales information, including special or mounted equipment. Sales information shall be limited to specifications and technical material, identical to that furnished to the authorized company representative for selection of truck models and components. Such items shall be construed as commercial motor vehicle products, and shall comply with all DoT Federal Motor Vehicle Safety Standards and Regulations applicable, on the date of manufacture. The chassis model furnished shall not be older than the chassis manufacturer's current model on the date of invitation for bids.

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3.3.1 Special requirements. In addition to standard truck and components specified in 3.3, truck shall be furnished with special equipment as specified herein.

3.3.1.1 Identification plate, data, and instruction plates.

* 3.3.1.1.1 Identification plate. Unless otherwise specified, an identification plate will be furnished by the contracting officer for each truck. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix it to each truck in a conspicuous place with nonferrous screws, rivets, or bolts not less than 1/8-inch in diameter. The applicable nomenclature shall be: TRUCK, AMMUNITION HANDLING, STAKE AND PLATFORM BODY.

3.3.1.1.2 Shifting instruction plate(s). A plate or plates showing applicable data concerning operating positions of shift handles for transmission transfer-case and power takeoff shall be furnished.

* 3.3.1.1.3 Crane instruction plates. Three crane operating instruction plates shall be furnished. A plate shall be located on the each side of the crane and shall be positioned to enable reading from ground level, and one shall be located inside the truck cab visible to the driver. The plates shall depict all necessary operating instructions to get the crane into an operating mode; for positioning (extending and/or retracting) outriggers; and for storage of crane before truck travel operation. Plates shall conform to composition A (class 1 or 2) or composition of MIL-P-514. The plates shall provide instructions as following:

* Inside cab plate.
a. Position truck as level as possible.
b. Apply and set parking brakes.
c. Start truck engine.
d. Engage power takeoff control.
e. Set throttle to recommended rpm's.
f. Follow instructions for operations of crane.
g. Release throttle high idle control.
h. Disengage PTO prior to operating truck.

* Plates located near crane control to include.
a. Lower and set outriggers.
b. Operate crane functions in accordance with manufacturer's instructions.
c. Stow crane before raising outriggers.
d. Raise and secure outriggers before moving truck.

3.3.1.2 Drain plugs. Drain plugs installed in manual transmissions, rear axle, and those installed for the purpose of draining crane and outrigger hydraulic fluid, shall be of the permanent magnet type.

3.3.1.3 Heavy-duty cooling system. A heavy-duty cooling system shall be furnished that shall maintain engine coolant at a temperature below the boiling point with truck loaded to rated gvw, and operated at an altitude of 10,000 feet

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above sea level or in an ambient temperature of not less than 125 degrees Fahrenheit (^{o}F).

3.3.1.4 Coolant system and indicators. Coolant system shall include a deaeration system, a surge tank or a coolant recovery reservoir of not less than 2 quart capacity. A low coolant level or high coolant temperature alarm buzzer and red indicator warning light shall be provided on the dash instrument panel.

3.3.1.5 Towing devices. Not less than two hooks or loops mounted to the frame or to rigid members which are attached to the frame, shall be furnished in front of the truck, for towing purposes.

* 3.3.1.6 Pintle assembly. A pintle assembly shall be furnished and shall conform to MS51335, part No. 51335-2. Rearmost portion of installed pintle shall be not more than 4 inches forward of the rearmost part of the vehicle. The attachment devices shall be capable of accommodating a standard grab hook (2-3/16 inches wide, 47/64 inch thick, and 31/64 inch throat width) for a 3/8 inch chain. When specified (see 6.2), the pintle height, measured to the centerline at the pintle, shall be 20 inches (+5,-0). Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. The pintle shall be installed with bracket reinforcements to transfer a vertical tongue load of not less than 1,800 pounds and a horizontal draw bar load of not less than 18,000 pounds directly to the chassis rails. The installation shall be capable of withstanding towed loads equal to the gvw of the truck without permanent deformation.

* 3.3.1.7 Splash guards and shields. Rear wheels shall have metal splash shields ahead of the wheels and rubber splash guards to the rear. Splash and stone throw protection shall be in accordance with SAE J682. A metal strip not less than 1/8 inch thick and not less than one wide, extending the entire width of the mud flap, shall be installed to prevent the bolts heads or bolts nuts from damaging the mud flap.

3.3.1.8 Radio interference suppression. The truck shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting electromagnetic radiation shall be suppressed to the same level as the vehicle chassis.

3.3.1.9 Air pollution control. Trucks shall comply with the EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, trucks destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

* 3.3.1.10 Sound level. The exterior sound level produced by the truck shall conform to the Environmental Protection Agency Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks. The interior sound level shall conform to MIL-STD-1474, Category D (85 db(A).

* 3.3.1.11 Lifting and tiedown attachments. When specified (see 6.2), the truck shall be equipped with lifting and tiedown attachments. Lifting and

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tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the truck. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tiedown of the truck on the carrier when shipped.

3.3.1.12 Engine hour meter. An engine hour meter, having a totalizing mechanism of not less than 9,999 hours, shall be furnished for the truck chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to ensure continuous trouble-free performance under severe operating conditions. Engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

* 3.3.1.13 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.3.1.14 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher. Modification to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Vehicle Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware.

* 3.3.1.15 4 by 4 conversion. When type II truck chassis is required, the chassis manufacturer's standard 4 by 2 truck chassis may be modified to provide all wheel drive requirements specified herein if:

(a) The conversion axle manufacturer's engineering department specifically approves and certifies that all such modifications meet the design requirements and standards of the conversion axle manufacturer. Certification shall be based on both design analysis and proving ground test reports, which shall be made available to the engineering and quality assurance offices of the procuring activity.

(b) The chassis manufacturer's front axle before conversion has the same load rating as the conversion axle to be installed.

(c) Components used in the all wheel drive conversion are of current production.

(d) Components used in the all wheel drive conversion are approved for the conversion application by the component manufacturers.

(e) The converted vehicle is certified to conform to Federal Motor Vehicle Safety Standard No. 121, by the intermediate or final manufacturer.

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(f) Replacement headlights, if required to be added to meet the height requirements of Federal Motor Vehicle Safety Standard No. 108, are equivalent in mounting, protection, and range and precision of adjustment to the chassis manufacturer's original standard headlights.

(g) Unused headlight cavities are covered in a neat workmanlike manner, treated and painted to match the chassis cab color with treatment and painting equivalent to the chassis cab manufacturer's process for the remainder of the chassis cab. Cavities and their covers shall be rustproofed in accordance with 3.14.

(h) Complete installation drawings are available to the procuring activity.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Weights, ratings and dimensions.

3.5.1 Curb weight. Curb weight shall include weight of chassis and cab, with all attachments, accessories, equipment, body, full complement of fuel, lubricants, and coolant.

3.5.2 Weight rating. The gvw shall consist of curb weight, operator (weight computed at 175 lbs), and payload to provide not less than the specified gvw.

3.5.3 Weight distribution. The distribution of the gvw for the purpose of establishing suspension, axle, and tire capacities shall be determined with the payload uniformly distributed over the load area.

* 3.5.4 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Components and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be submitted to the engineering office of the procuring activity. The minimum GVW rating shall be 24,000 pounds.

3.5.5 Cab-to-axle. The cab-to-axle dimension shall be not less than 108 inches.

3.5.6 Overall width. The overall width of the vehicle, exclusive of mirrors, lights, reflectors and tires, shall be not more than 96 inches. The width over the tires shall be not more than 100 inches.

3.6 Speeds and gradeability. High and low speed requirements shall be met with truck loaded to specified gvw.

3.6.1 High speed gradeability. Truck shall ascend a continuous 1.0 percent grade at 50 miles per hour (mph). Gradeability requirement shall be met with the main transmission in direct drive, transfer case in high range and with the multispeed axle, auxiliary transmission or transfer case in high speed range. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.4).

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3.6.1.1 Low speed. Low speed, with engine operating at not less than 35 percent of recommended governed speed, shall provide a truck speed of not more than 5 mph.

3.6.1.2 Maximum geared speed. Maximum geared speed at engine governed speed shall be not less that 58 mph.

3.6.2 Brake performance. Service brakes shall comply with the performance requirements specified in DoT Federal Motor Carrier Safety Regulations, section 393.52. Service brakes shall control and hold the truck, loaded to the specified gvw, on a 30 percent grade. Parking brake shall hold the truck, loaded to the specified gvw, headed either up or down the grade, without slipping on a 30 percent grade.

3.7 Chassis components.

3.7.1 Engine. The engine furnished shall be the chassis manufacturer's standard, or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

* 3.7.1.1 Gasoline engine. When specified (see 3.7.1.2), the truck shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine with not less than six cylinders. The engine furnished shall produce the required truck performance using unleaded gasoline with a research octane rating of 91 at a speed not more than the manufacturer's recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel when operated in accordance with the operator's manual. Engine net horsepower (hp) used in performance prediction calculations shall be determined in accordance with SAE J1349.

* 3.7.1.2 Diesel engine. Unless otherwise specified (see 6.2), the truck shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine with not less than six cylinders. Engine net hp used in performance prediction calculations shall be determined in accordance with SAE J1349. A fan clutch to reduce fan speed automatically, when not required for engine cooling, shall be provided. The fan clutch shall be asbestos free.

3.7.1.3 Oil filter. Full flow type oil filter shall be furnished.

3.7.1.4 Governor. Engine governor shall be set and sealed to limit engine to the engine manufacturer's maximum recommended operating speed.

3.7.2 Electrical system.

3.7.2.1 Ignition system. For gasoline engine driven trucks, the ignition system shall be 12 volts (V) potential, negative ground. Alternator of not less than 50 ampere rated capacity, and which provides not less than 16 amperes direct current output at normal engine idle speed, shall be furnished. The entire exposed wiring system shall be normally resistant to moisture and fungi without treatment.

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3.7.2.2 Starting system. For diesel engine driven trucks , a 12- or 24-volt starting system, with 12-volt direct current lighting system, and not less than a 60 ampere alternator, shall be furnished. Engine starting equipment shall include an ether starting system or glow plug. When an ether starting system is furnished, it shall be of the measured shot type, key operated or manually controlled from the driver's compartment and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 12 ounces shall be furnished. A reservoir need not be furnished.

3.7.2.3 Battery. Each battery shall be 12V potential. For gasoline engine driven trucks, the total reserve capacity rating shall be not less than 100 minutes and the total cold cranking ampere rating at 0°F shall be not less than 450 amperes. For diesel engine driven trucks, the total reserve capacity rating shall be not less than 320 minutes and the total cold cranking ampere rating at 0°F shall be not less than 1,200 amperes. The batteries shall be of the maintenance free type with the maintenance free characteristics listed in W-B-131.

* 3.7.2.4 Lighting. All truck lights, reflectors, and wiring shall be as specified herein and shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.12, 393.19 through 393.31, and 393.33, as applicable. Lights and reflectors shall not be mounted on vertical surfaces of rub rails (unless recessed and fully protected) or mounted on bumpers. Brake lights shall override the emergency flashers on this vehicle.

3.7.2.5 Blackout lights. When specified (see 6.2), in addition to lighting specified in 3.7.2.4, trucks shall be provided with combination blackout lights, taillights, and stoplights and shall be in accordance with MS51318, MS52125, and MS52126. Blackout lights shall be wired in such a manner as to preclude inadvertent lighting of the service lights when blackout lights are turned on, except for service stoplights which shall function during all operating conditions when brakes are applied. Blackout lights shall be installed at front and rear of truck. Those at the front shall be as far apart as practicable. Those at the rear shall be in the same housing as stop and tail lamps. The headlight blackout light, conforming to MS51318, shall be located on the left side front of the truck, and the blackout marker lights, conforming to MS52126-1, shall be located on the right and left side on the front of the truck The rear blackout lights, conforming to MS52125-1, shall be located, one on each side, at the rear of the truck. All lighting shall be equipped with 12V bulbs.

3.7.2.6 Trailer electrical connector. A seven connector trailer electrical connector and cover assembly, conforming to SAE J560, with conductors connected and color coded as specified therein, shall be provided at a readily accessible location on the rear of the truck.

3.7.2.7 Electromagnetic compatibility requirement. When specified (see 6.2) the trucks shall conform to electromagnetic compatibility requirements in accordance with MIL-E-6051.

3.7.3 Fuel system. Fuel system shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.65 and 393.67.

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3.7.3.1 Air cleaner. A dry type air cleaner shall be furnished.

3.7.3.2 Fuel tank(s). Fuel tank(s) shall be manufacturer's standard for the truck furnished. Fuel capacity shall be not less then 43 gallons. When more than one tank is furnished on diesel engine driven trucks, means shall be provided to assure equalized fuel level in both tanks. When more than one fuel tank is furnished on gasoline engine driven trucks, a selector valve connecting either fuel tank to the engine fuel intake shall be provided and means shall be provided to monitor the fuel level of either tank from a single fuel gage.

3.7.4 Exhaust system. Exhaust system conform to DoT Federal Motor Carrier Safety Regulations, section 393.83. A spark arrester shall be furnished on vehicles, except with turbocharged engines. The spark arrester shall have an 80 percent arresting efficiency when rated in accordance with SAE J350. A shield plate shall be installed, where necessary, to prevent spilled fuel and lubricants from coming in contact with the exhaust system. When furnished vertical exhaust mufflers capable of being reached easily by personnel entering or leaving either side of the cab shall be provided with a heat shield. When furnished, vertical exhaust system shall be provided with a hinged rain cap.

* 3.7.5. Automatic or semiautomatic transmission. Unless otherwise specified, (see 3.7.6), the vehicle shall be provided with an automatic or semiautomatic transmission. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The automatic or semiautomatic transmission furnished shall be of the fixed ratio type and shall include a hydraulic torque converter and not less than five forward gear ratios. Normal driving range selector position shall be provide not less than four gear ratios without movement of the selector. The transmission shall be provided with an SAE J704 power takeoff opening.

3.7.6 Transmission. When specified (see 6.2), a manually shifted five forward speed main transmission shall be furnished. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. Gear ratios in transmission, auxiliary transmission or transfer case (when furnished) and rear axle shall be matched to provide a progressive shifting pattern throughout the complete range and shall provide the required truck performance (see 3.6.1). Transmission shall be provided with SAE J704 power takeoff openings at right and left sides. The transmission shall provide for maximum ease of shifting at all speeds. A positive, manually operated, securing device shall be provided to lock the truck transmission to prevent transmission of power to the truck wheels when the power takeoff is engaged.

3.7.6.1 Speeds. The vehicle shall be provided with not less than ten forward speeds. The 4 by 2 model vehicle shall have an auxiliary transmission or 2-speed rear axle. The 4 by 4 model vehicle shall have a 2-speed transfer case and a single-speed rear axle. Gear ratios in the transmission or other gear selection equipment shall be matched to provide a progressive shifting pattern throughout the complete range and shall provide the required vehicle performance specified in 3.6.1

* 3.7.6.2 Transfer case (model 4 by 4). Unless the transfer case is equipped with devices which compensate for differential torque and speeds between the

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front and rear axles, the transfer case shall provide for driver selection of either two-wheel or four-wheel drive. When interaxle compensating devices are furnished, they shall provide for the positive transfer of power to all the driving axles. A two speed transfer case shall be furnished. The vehicle speedometer shall read accurate speed (mph) with the speed selector in high or low range.

* 3.7.6.3 Clutch. Clutch shall be the largest capacity optional clutch offered for the truck chassis and engine furnished, with torque capacity exceeding the maximum delivered engine torque. The clutch shall not contain any asbestos material.

3.7.7 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

* 3.7.8 Frame. A chassis frame with reinforcement shall be furnished. The frame reinforcement shall extend at least from the rear of the front suspension, rear hanger bracket to at least the edge adjacent to the rear spring, front hanger bracket. The chassis frame with reinforcement shall withstand all loads imposed by the crane (see 3.8) with the crane loaded to rated capacity in all its attainable positions.

* 3.7.9 Suspension. Truck shall be equipped with suspension system with components having a minimum rated capacity equal to the load imposed on each member, measured at the ground, with truck loaded to rated gvw. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted. The chassis shall be equipped with hydraulic double-acting shock absorbers at the front wheels.

* 3.7.10 Axles. Minimum axle ratings shall be equal to loads imposed on each axle, measured at the ground, with truck loaded to maximum rated gvw, and of such gear ratios to meet performance requirements of 3.6.1. When a two speed drive axle is furnished on 4 by 2 chassis, it shall be operated by electric, vacuum powered , or air shift device. A single speed rear drive axle shall be furnished on 4 by 4 chassis.

* 3.7.10.1 Traction control. The rear axle shall be furnished with furnished with an increased traction device. Maximum traction capabilities shall be maintained at all times under each drive wheel during operation.

* 3.7.11 Wheels, rims, tires, and tubes. Truck shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to TRA recommendations, for the type and size furnished. Tire and rim sizes shall be the same for all wheels on each truck. Disk type wheels shall be furnished. Multi-piece rims shall not be used in this vehicle.

3.7.11.1 Tires. Tires shall be tube or tubeless type with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with truck loaded to rated gvw. Steel belted radial tires are not acceptable. Tires shall conform to Tire and

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Rim Association recommendations or to ZZ-T-381 with a tire designation system the same as the Tire and Rim Association.

3.7.11.2 Tubes. When tube-type tires are furnished, tubes shall be of heavy-duty type, and shall be of proper size for tires furnished. Tire flaps shall be provided for tube type tires in accordance with TRA recommendations.

3.7.11.3 Spare wheel assembly. A spare wheel shall be mounted on a carrier and installed in a readily accessible location on the truck.

3.7.11.4 Tire chain clearance. Tire chain clearance shall be in accordance with SAE J683. Allowance for spring deflection shall be included.

3.7.12 Brakes. Brakes shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.40 through 393.42, and 393.45 through 393.52. Brake linings shall be of nonasbestos material.

* 3.7.12.1 Air brakes. When specified (see 6.2), the trucks shall be equipped with full air brakes on all wheels. The braking system, complete with all necessary components, shall include:

a. Air compressor, unloader-head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less the 7-1/4 cubic feet per minute (cfm).
b. Air storage reservoir(s) with not less than 2,000 cubic inches total capacity, each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank.
c. Foot control, suspended or treadle type.
d. Air control valves.
e. Air pressure gage, visible to the driver.
f. Low air pressure warning, visible and audible.
g. Service brake stop lamp switch.
h. Alcohol aspirator with unbreakable transparent container.
i. Automatic moisture ejector.
j. When specified (see 6.2), automatic slack adjusters.

* 3.7.12.1.1 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

* 3.7.13 Cab. Manufacturer shall furnish a standard full width model cab. Cab doors shall be equipped with locks, operable from inside the cab and with at least one door equipped with external, key-operated lock. Drip rails shall be provided above the cab doors. Interior dome lighting shall be provided. Manufacturer's standard full width seat and three pairs of seat belts shall be furnished on the truck. The color of the upholstery and interior finish shall be compatible with the exterior color (see 3.13). White upholstery shall not be furnished.

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* 3.7.14 Heater and defroster. Manufacturer's standard hot water heater and defroster shall be provided. The system shall have include fresh air intakes and discharge areas for floor and windshield defrosting. Heater and defroster controls shall be convenient to the operator's position.

* 3.7.15 Windshield wipers and washers. Truck shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type and operated by either air or electric motor(s). When electric powered wipers are furnished, controls shall include intermittent operation mode.

* 3.7.16 Steering. Manufacturer's standard power assisted steering shall be furnished.

3.7.17 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of operator to and from driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be panel mounted.

* 3.7.18 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following equipment as a minimum shall be furnished:

a. Combination (flat and convex) outside rearview mirrors shall be provided on each side of cab. Each combination mirror shall have at least 50 square inches of flat reflective area and a convex surface having at least 25 square inches of reflective area. Each mirror shall have at least two supporting arms.
b. Key-operated ignition switch.
c. Ammeter or voltmeter charging indicator.
d. Fuel gage(s).
e. Oil pressure gage.
f. Engine temperature gage.
g. Speedometer with recording odometer.
h. Dual sun visors.
i. Door mounted armrest on driver's side.
j. Driver's compartment ventilator other than window.
k. Manufacturer's standard horn.
l. Adjustable throttle, for hoist operations.
m. Audible backup alarm device.

* 3.7.19 Front bumper. Manufacturer's standard front bumper shall be furnished.

3.7.20 Rear end protection. Rear end of the truck shall be protected in accordance with DoT Federal Motor Carrier Safety Regulations, section 393.86. Rear bumper or bumperettes shall be furnished.

* 3.7.21 Front mounted winch. When specified (see 6.2), truck shall be furnished with a winch assembly, mounted on the front of the truck and powered

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by manufacturer's standard power takeoff. The winch shall conform to SAE J706. Winch shall be single drum type and shall have a minimum one forward and one reverse speed. All winch controls shall be inside cab mounted. Winch controls shall be located to provide no interference with the ingress or egress of the driver. Winch shall have a rated single line pull capacity of not less than 20,000 lbs on the bare drum. Winch shall be wound with not less than 185 feet of 5/8-inch diameter cable of improved plow steel, wire strand core, and shall be equipped with end chain and hook. An integral, adjustable, automatic safety brake shall be provided. Winch shall be equipped with a roller guide. Angles of the winch drive line U-joints shall be not greater than 17°.

* 3.7.21.1 Winch drum guard. Winch drum guard shall be furnished. The guard shall confine the cable to the area between the drum flanges.

* 3.7.22 Angle of approach. The angle of approach shall be not less that 22°, with or without winch assembly, when the truck is loaded to its maximum GVW.

3.7.23 Fire extinguishers and brackets. Two fire extinguishers and two brackets for retaining the fire extinguishers shall be furnished. Fire extinguishers shall be in accordance with A-A-393, type II, class 2, size 10. The fire extinguisher shutoff valve shall be located at the outlet of the discharge hose. Design of each bracket shall enable extinguisher to be readily released in one operation with one hand. Brackets shall be mounted one on each side of the truck. Adequate protection from splashing of mud and water shall be provided. Fire extinguishers shall be located as to be readily accessible to anyone standing on the ground.

3.7.24 Rotating warning light. An amber, rotating, vehicular warning light shall be furnished and shall conform to MS51317, part No. 51317-4. Rotating light shall be mounted with reinforcement, at point of attachment on truck centerline for 360° visibility. Rotating light shall be operated by means of a separate switch with indicator light located on cab instrument panel, and marked to indicate function and position of switch.

* 3.8 Crane. A one-man operated, fully hydraulic, articulated boom type crane shall be mounted on the truck chassis frame between the cab and platform body. The maximum space behind the cab for crane installation shall be 28 inches. The crane shall be powered by transmission mounted power takeoff. The boom shall consist of an upper and lower section and include a hydraulically operated extendable jib. The total extension and retraction of the jib shall be hydraulically controlled. The boom, when fully extended, shall reach not less than 18 feet and hoist not less than 23 feet above ground level, and shall rotate safely through not less that 190° with no drift rotation. The boom shall fold to a nestable travel height between platform body and truck cab, and shall be not more than 85 inches above truck chassis frame. A means shall be provided to protect the truck cab from damage due to crane rotation during stowage positioning and crane swing operations. The crane design shall provide for smooth and quiet operation, ease and flexibility of operation, and versatility of performance. Crane shall be provided with load holding capability that will enable the crane under static load conditions to maintain zero vertical downward

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creeping of the load. The truck mounted crane shall comply with Safety Standards for Mobile Hydraulic Cranes, ANSI B30.5.

3.8.1 Crane capacity. With outriggers in a down position, the crane shall be capable of lifting a minimum load of not less than 4,000 lbs, when the load is positioned at a horizontal working radius of 10 feet, and a minimum load of not less than 2,000 lbs at a horizontal working radius of 18 feet. The crane horizontal working radius is the measured distance from centerline of crane rotation to the centerline of the lifting hook. A capacity chart shall be provided showing as a minimum the maximum allowable loads at working radii of 18 feet, 16 feet, 14 feet, and 10 feet. The crane capacity chart shall be provided and shall be located one on each side of the truck, in a position to enable the crane operator to read it from ground level.

3.8.2 Crane hoist hook. Unless otherwise specified (see 6.2), manufacturer's standard hoist hook shall be provided, furnished with a swivel safety hook, and shall have not less than a 2-inch throat opening. The rated capacity of the crane hook shall be not less than the rated capacity of the crane for which it is furnished. The hook shall be of the forged steel, swivel type with an ultimate strength of four-times the maximum safe-rated load and shall include a safety latch on the hook. The block and hook shall have adequate weight to ensure payout of the cable without load on the hook in any position. Block assembly shall be provided for attachment to the hoist cable for not less than two-part purchase. Means shall be provided to secure hook block while traveling. The hook block furnished shall permit disassembly for visual and nondestructive testing. The hook shall not be painted.

3.8.3 Outriggers. Double-acting hydraulic outriggers, independently controlled and integrally mounted to the crane base, shall be furnished. Two outriggers shall be provided, one on each side of the truck. Each outrigger landing pad shall be not less than 140 square inches. Pads shall be pinned to outrigger if removal is necessary. The outriggers shall provide side-to-side leveling of the truck on inclined surfaces and on level surfaces. The overall length of the fully retracted outrigger shall be not greater than the width of the platform body.

3.8.3.1 Outrigger warning light. A red warning light shall be located and installed in the cab in a conspicuous location to warn the driver that the outriggers are not fully retracted. The warning light shall be suitably labeled to indicate light function.

3.8.4 Crane mounting. Unless otherwise specified (see 6.2), the mounting of the crane on the truck chassis frame shall be the manufacturer's standard method.

3.8.5 Hydraulic system. Hydraulic system with power takeoff, pump, piping, valves, controls, and other components necessary for a complete installation, shall be furnished for the crane and outriggers. Power takeoff for the hydraulic pump shall be controlled from the inside of the truck cab. A hydraulic fluid filter and a reservoir shall be provided. The hydraulic system shall contain the following minimum features:

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a. Check valve system in boom and outrigger system to prevent load drop due to hydraulic system failure.
b. Constant flow valve in hydraulic lines to preclude boom damage due to sudden load lowering stops.
c. Cushioning valves in boom rotating hydraulic system to preclude damage due to sudden stops or extreme induced side loads.
d. A pressure relief valve shall be provided to prevent damage to pump, valves, and hoses when the hydraulic system is over pressurized.
e. Signs with 1-inch lettering, visible to the operator at both sets of crane controls: OUTRIGGERS MUST BE FULLY EXTENDED AND IN PLACE BEFORE LIFT IS MADE.
f. Each safety check pilot valve, flow valves, cylinder, and hydraulic lines shall be free of contaminates.
g. Clearly identify power takeoff lever in cab for engaging crane hydraulic pump.
h. Provide positive seating or holding valves that will enable crane, under static load condition, to hold maximum load without vertical creep due to hydraulic system malfunction or failure.
i. A red warning light shall be located and installed in the cab in a conspicuous location to warn the driver that the power takeoff unit is engaged. The warning light shall be suitably labeled to indicate light function. A caution plate reading: "Do not operate vehicle at highway speeds with power takeoff engaged" shall be installed.

3.8.6 Controls. Crane controls shall be provided on each side of the truck. Each outrigger control shall be located on the same side of the truck as the outrigger. Crane and outrigger controls shall be accessible from ground level. Controls shall be of the deadman type to ensure automatic return to neutral position in event of either inadvertent or intentional release by the operator. All crane and outrigger controls shall be identified to indicate what it is and how it is to be used. Provisions to ensure against engagement of power to truck wheels and crane at the same time shall be provided. Controls shall have fine metering qualities to provide infinitely variable raising, lowering, and rotating motions of the crane boom. There shall be no acceleration or surging of the load when activating crane control.

3.8.7 Stability warning device. An audible warning device shall be provided that will alert the crane operator when swinging the load, that a potentially unstable truck position exists either caused by overload or partially extended outrigger. This device shall be installed in conjunction with power takeoff operation and shall activate from 5° off the truck horizontal plane.

3.8.7.1 Overload shutdown system. The crane shall be provided with an overload shutdown system that shall preclude structurally overloading the crane. All crane functions shall shut off when the crane's capacity is exceeded by 30 percent during such operations of load lowering and/or boom extension of the load that would increase the crane's working radius. A manual override, of the deadman type, shall be provided to return the crane to normal operation after the overload shutdown system has been activated. The crane overload system

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shall be designed fail-safe operation in event the crane's electrical or hydraulic system should become inoperative. Truck engine shutdown when system is activated is not acceptable.

3.8.8 Environmental performance. The crane shall perform satisfactorily all requirements specified herein in any ambient temperature from -25°F to +125°F without any malfunction.

3.8.9 Stress requirements. Design of the truck mounted crane shall be such that the sustained maximum unit stress imposed by the rated load shall be not more than the minimum strength margins established in SAE J1063 for cantilevered boom crane structures. When specified (see 6.2), a strain gage test shall be performed on the first production vehicle in accordance with the methods and procedures set forth in SAE J1063.

* 3.9 Body. The truck shall be equipped with an all steel constructed platform type body. The overall length of body shall be not less than 144 inches. The overall width of body shall be not more than 96 inches. The platform body frame shall be provided with longitudinal members and crossmembers of sufficient strength to support the payload (see 3.5.2 through 3.5.4). The platform body shall be provided with sufficient stake pockets to accommodate the removable type body front and side panels specified herein (see 3.9.1 through 3.9.1.2). The stake sides and pockets shall include retention devices to prevent accidental detachment. The devices shall not require tools to remove stake sides. Unless otherwise specified (see 6.2), the platform body shall be mounted on the truck chassis frame with the manufacturer's standard method. The floor of the platform body shall be steel, of sufficient thickness to support a concentrated load of 300 pounds per square foot at any point on the floor load space. The steel platform deck shall be covered with 3/4-inch exterior marine grade plywood fastened with countersunk aluminum bolts. Effective means shall be provided to impede electrolytic/galvanic corrosion between nonferrous fasteners and steel floor. When specified (see 6.2), the platform body shall be mounted on the truck chassis frame with quick-disconnect devices incorporating quick-release bolts so body can be quickly removed from truck chassis for helicopter lift as a separate unit.

3.9.1 Cargo conversion unit. Unless otherwise specified (see 6.2), a cargo conversion unit shall be furnished and shall consist of body front, sides, tailgate, apertures, and cargo ties as specified herein. The body front, sides, and tailgate shall be fabricated from steel, and provide a smooth inside surface. The stakes of the body sides and front panels shall face to the exterior of the body. The body panels shall face toward the load space. The tailgate panel shall face toward the body load space. The body panels and tailgate shall be of not less than 0.0478-inch (US revised standard gage No. 18) thick sheet steel.

3.9.1.1 Body front panel. The body front panel shall be not less than 14 inches in height and shall be provided with not less than three stakes. The top and bottom of the panel shall be formed outward into a box section of sufficient width to provide stake pockets. The body panel stakes shall be not less than 3 by 2 rectangular steel section of not less than 1/8 inch and shall be attached to the outside of the panel through the stake pockets. The body

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front panel shall be secured to the platform using the same retention devices as the side stakes.

3.9.1.2 Body side panels. The body side panels shall be not less than 14 inches in height. Three body side panels shall be provided on each side of the platform body. Each body side panel shall be provided with not less than two stakes. The body panel stakes shall be not less than 3 by 2 rectangular steel section of not less than 1/8 inch, and shall be attached to the outside of the panel. The stake length shall be such as to be attached for the full height of the body panel plus the required length to allow attachment into stake pocket. Each forward side panel shall be attached to the front panel. Each rearward side panel shall be attached to the intermediate side panel. Each side and end panel shall be furnished with attachment devices to lock to each other. These devices shall not require tools to operate, and shall allow individual panel removal.

3.9.1.3 Tailgate. The tailgate shall be not less than 16 inches in height. The tailgate shall be flanged outward into a box section on top and sides and reinforced on the bottom edge. The tailgate shall be hinged at the bottom with four hinges; the upper hinge strap shall run the full height of the tailgate. Removable type fastenings shall be provided for the tailgate hinges that are secured to the platform body. Tailgate locking devices shall be provided. Tailgate shall be capable of traversing 180° from the vertical, closed position. Tailgate shall be provided with duck covered stay-chains of a length to retain the tailgate at a horizontal position. When tailgate is lowered to the horizontal position, the tailgate shall be flush with the platform body. Stirrup steps, of 200 lbs capacity each, shall be attached on the top flange on each side of the tailgate to assist personnel in mounting and dismounting. The tailgate assembly shall support, without deformation, a total of not less than 400 lbs. Tailgate shall not interfere with pintle hook assembly. Rear lights and reflectors shall be positioned to preclude damage by lowering of gate.

3.9.1.4 Apertures. The platform body floor shall be provided with 24 apertures. Each aperture shall be 2-1/2 inches in diameter. The apertures shall be arranged in the following pattern: 1 foot from each platform edge, and 2 feet between centerline of apertures. The apertures shall be deburred to preclude the edges from cutting the cargo ties.

3.9.1.5 Cargo ties. Eight sets of cargo ties shall be furnished for each truck. Each set shall consist of two units. One unit shall be not less than 3 feet long webbing having a buckle secured on one end and a pin permanently attached to the other end. The second unit shall be not less than 15 feet long webbing having a pin permanently attached to one end, and the other end shall have a metal protective device to prevent fraying. The webbing shall be 1-3/4 inches wide, fire, water, weather, and mildew resistant nylon. The buckle shall be 1-3/4 heavy duty, corrosion-resistant ratchet type. The pin shall be not less than 3/8-inch round steel bar stock 4-1/2 inches long. The total assembly of the set consisting of the two units, shall have a tensile strength of not less than 5,000 lbs. Storage space shall be provided inside the driver's compartment for the eight sets of cargo ties.

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3.10 Floodlights. Two, 6-inch minimum diameter, adjustable floodlights shall be installed, one on each rear corner of the cab, to illuminate crane working area and platform body area. Each floodlight shall be provided with a weather-resistant toggle switch.

* 3.11 Tool compartment. A weatherproof tool compartment of not less that 8 cubic feet capacity shall be furnished. The tool compartment shall be installed at the forward end of the body under the chassis at the curb side. Provisions for locking with padlock shall be provided.

3.12 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 pound-force per square inch or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.13 Treatment, painting, identification marking. As specified by the procuring activity for the appropriate service (see 6.2), treatment, painting, and identification marking shall be in accordance with MIL-STD-1223.

3.14 Rustproofing. When specified (see 6.2), the truck cab, chassis frame, and chassis underside shall be rustproofed in accordance with FED-STD-297.

3.15 Servicing and adjusting. Prior to acceptance of the truck by the Government, the contractor shall service and adjust the truck for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes (when required).
c. Proper functioning of all lighting and electrical systems.
d. Wheel alignment (when required).
e. Adjustment of engine to include tune-up.
f. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
g. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

The truck shall be conspicuously tagged to identify the lubricants and their temperature range.

3.16 Workmanship.

3.16.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

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3.16.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.16.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened shall be of approved shape and of uniform size for the same diameter or rivet. The rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.16.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

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4.2.1 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. This inspection shall include the examination of 4.3 and the test of 4.4. The purpose of the inspection shall be to determine vehicle conformity with the requirements of the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its right under the provisions of the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.5, and the packaging inspection of 4.6.

4.3 Examination. Each truck shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 First production vehicle test. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformity with the requirements of the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its right under the provisions of the contract.

4.4.1 Vehicle weight. The truck shall be weighed to determine curb weight and distribution of curb weight on front and rear axle. The imposed loading on front and rear axle will be computed using the curb weight, the operator weight at 175 lbs, and the payload (see 3.5.2), to provide the specified gvw. The calculated imposed loads on front and rear axle will be utilized to ascertain that the suspension, axles, and tires furnished are of adequate capacity to meet the requirements of 3.7.9, 3.7.10, and 3.7.11.1.

* 4.4.2 Road test (Type I, 4 by 2 truck). The type I truck, loaded to the specified gvw, shall be driven at a speed of 50 mph over improved roads for not less than 30 miles, and at a speed of 20 mph over unimproved roads for not less than 20 miles. The truck, loaded to the specified gvw, shall be driven at speeds varying from 5 mph to 20 mph, acceleration and deceleration operation, for not less than 10 miles. Upon completion of the test, the truck and equipment shall be inspected for leaks, warping, distortion, or malfunction.

* 4.4.2.1 Road test (type II 4 by 4 truck). Type II truck for models built to 4 wheel drive while on the chassis manufacturer's production line shall be road tested by the contractor with payload to provide specified GVW. The road test shall be for not less than 30 miles at speeds up to 55 mph, and for not less than 20 miles at a speed of 20 mph. The vehicle, for 4 wheel drive models qualifying under 3.1.1.15, shall be road tested with and without payload.

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Payload shall be distributed first, so that the front axle is loaded to GAWR and the vehicle is loaded to GVW. The road test for each of the three conditions shall be not less than 30 miles at speeds up to 50 mph over highways and gravel roads and for not less than 5 miles at speeds up to 15 mph over cross country terrain with ground and grade requiring all wheel drive. During the loaded and empty portions of the road test, the brakes shall be applied firmly, bringing the vehicle to a sudden stop not less than 5 times during each portion of the road test. During the road test, the front and rear suspension and the drivetrain shall be periodically inspected for interference and contact with other vehicle components. Abnormal contact of the drivetrain or suspension components with other components shall be cause for rejection. Front spring bumpers shall not make contact with frame stops except under the most extreme cross country conditions. Operational or mechanical failures of vehicle components during the road test shall be cause of rejection. Failure includes permanent deformation as well as breakage.

* 4.4.2.2 Documentation. The contractor shall furnish to the Government, copies of the certification and test reports required by 3.1.1.15(a) for Type II vehicles with 4 by 4 conversion.

4.4.3 Service brake test. The truck shall be loaded to rated capacity. The truck shall be subjected to three emergency stops in rapid succession from 20 mph on clean, dry, smooth, level, paved surface free from loose material. Each stop shall be accomplished in not more than 1-minute intervals. The truck shall stop within 40 feet from the point the brakes are applied. This test shall be conducted not less than four times over the duration of the road test.

4.4.4 Crane operation test. The crane shall be operated through at least 10 cycles of loading and unloading truck at each safe rated load capacity and radii to verify conformance to the capacity requirements of 3.8.1. During the cycle tests, the following shall be accomplished:

a. The rotational and lifting requirements shall be verified to determine conformance to 3.8.
b. Verify that hydraulic system safety features and devices conform to requirements of 3.8.5.
c. Verify that crane controls conform to the requirements of 3.8.6.
d. Verify that the stability warning device conforms to the requirements of 3.8.7.
e. Verify that the overload shutdown system conforms to the requirements of 3.8.7.1.

4.4.5 Static overload test. Static overload test shall be performed on the crane with 110 percent of rated capacity at the minimum and maximum rated radii.

4.4.6 Vertical creep test. Verify conformance to 3.8.5(h) as follows:

a. With the outriggers set, boom at minimum and maximum extension positioned approximately horizontal in the most stable direction, lift a rated load, and hold the rated load for 10 minutes. Without use of controls by the operator, there shall be no lowering of the load, boom, or outriggers due to hydraulic leakage, valve bypass,

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system or component malfunction or failure for the duration of the test. A copy of the test report of this verification shall be provided with each truck shipped.

4.4.7 Environmental test. The truck mounted crane shall be tested to verify conformance to operational capabilities at low and high temperature requirements of 3.8.8. After a minimum 8-hour soak period at each temperature range, the crane shall be operated under load through all crane functions. The truck cab heater and defroster shall be checked during low temperature tests. To determine conformance to 3.8.8, the manufacturer may certify that a complete test has been conducted on an identical truck mounted crane model and test records are available for inspection. If the certification is not furnished, and the test records are not available, then the test shall be conducted as specified herein.

4.4.8 Crane stress requirement test. When crane strain gage stress requirement verification test is required, the truck mounted crane shall be tested in accordance with SAE J1063 as set forth in 3.8.9. To determine conformance to 3.8.9, the manufacturer may certify that a complete test has been conducted on an identical truck mounted crane model and test records are available for inspection. If this certification is not furnished and test records are not available, then the test shall be conducted in accordance with SAE J1063.

4.4.9 Lifting attachment test. When lifting attachments are required, the components shall be suspended by the lifting attachments with suitable slings. The load on each lifting attachment shall be determined with the component in suspension. The component shall be lowered and each lifting attachment shall be subjected to a pull equal to or greater than 2-1/2 times the load reading previously obtained on the same lifting attachment while the component was suspended to determine conformance to 3.3.1.11.

4.4.10 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under the contract until corrective action has been taken. 4.4.11 production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and may be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.4.11 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and may be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.5 Production truck test. Each production truck shall be tested as specified in 4.4.4 (a, b, c, d, and e), 4.4.5 and 4.4.6.

4.6 Packaging inspection. The vehicle shall be inspected to verify conformance to the requirements of section 5.

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5. PACKAGING

* 5.1 Vehicle processing. The equipment shall be preserved and packed in accordance with the contractor's standard practice. When specified (see 6.2), equipment shall be preserved, packed, and marked in accordance with the mobile requirements of MIL-V-62038 with the level of preservation and packing as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The trucks covered by this specification are intended for use in loading, unloading, and transporting munitions at military installations.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type of truck chassis required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When pintle height of 20 inches (+5,-0), is required (see 3.3.1.6).
e. When lifting attachments are required (see 3.3.1.11).
f. When gasoline engine is required (see 3.7.1.2).
g. When blackout lights are required (see 3.7.2.5).
h. When electromagnetic compatibility requirement is required (see 3.7.2.7)
i. When manual shift transmission is required (see 3.7.6).
j. When automatic slack adjusters are required (see 3.7.12.1).
k. When front mounted winch is required (see 3.7.21).
l. When crane hoist hook other than specified is required (see 3.8.2).
m. When mounting of the crane is not manufacturer's standard method (see 3.8.4).
n. When crane strain gage stress requirement verification test is required (see 3.8.9).
o. When mounting of the platform body is manufacturer's standard method and when quick-disconnect type platform body mounting devices are required (see 3.9).
p. When cargo conversion body unit is not required (see 3.9.1).
q. Appropriate service for treatment, painting, and identification marking (see 3.13).
r. When rustproofing is required (see 3.14).
s. When packaging in accordance with MIL-V-62038 is required and the level of preservation/packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL),

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incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 Performance prediction. Completed copies of SAE Truck Ability Prediction Procedure form and computation for speed requirements for each vehicle model furnished under contract should be submitted as specified in the contract. The SAE Work Sheet Item should include vehicle model number, wheelbase, engine model number, and vehicle type. Unless other conditions are sighted in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature and still, dry air. The factors to be used in predicting truck ability (see 3.6.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1	-	Tire factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.	
Table 2	-	Altitude factor.	1.00
Table 3	-	Rolling factor.	1.613
Table 4	-	Area factor. Use factor.	0.1173
Table 5	-	Velocity factor.	250.0
Table 6	-	Altitude factor.	1.00
Table 7	-	Chassis friction hp. Use applicable power unit gvw (to nearest, high 1,000 lbs) and the engine 100 revolutions which is required for 50 mph geared speed. For gvw and engine speed beyond the range of this table, factors shall be extrapolated.	
Table 8	-	Grade factor.	0.75
Table 8A	-	Correction factor.	Not required
Table 9	-	Road factor.	0.0

6.5 Subject term (key word) listing.

Ammunition
Diesel engine
Platform
Truck

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AT
Navy - YD

Preparing Activity:
Navy - YD

(Project 2320-0444)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-81098F	91-02-20

3. DOCUMENT TITLE

TRUCK, AMMUNITION HANDLING: STAKE AND PLATFORM BODY, 24,000 POUNDS GVW, 4BY2 OR 4BY4

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial: (805) 982-5572 (2) AUTOVON: 551-5572
c. ADDRESS (Include Zip Code) Commanding Officer (156) Naval Construction Battalion Center Port Hueneme, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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